

Decision Memo: Drone Depot Swap Capacity

To: Operations Manager, Bosphorus Drone Light Festival
From: IE 306 Analytics Team
Date: May 31, 2026
Re: Whether to purchase one additional battery-swap station

Recommendation

Purchase the sixth swap station, provided its capital cost is reasonable relative to the value of reduced depot delay. The simulation evidence shows a clear and practically meaningful reduction in steady-state swap-queue wait.

Evidence

I treated the depot as a steady-state system and deleted the initial synchronized-start transient. Welch's method suggested 7,200 seconds, MSER suggested 7,620 seconds, and I used 7,620 seconds downstream. Under Policy A, replications-with-deletion gives a mean swap-queue wait of 14.46 seconds with 95% half-width 0.48 seconds. Batch means gives 15.02 seconds with half-width 0.60 seconds using $B=200$, $K=492$, and lag-1 correlation 0.083; the intervals overlap.

Policy Comparison

Using common random numbers and 50 paired replications, the estimated improvement is 10.35 seconds per drone landing, measured as Policy A minus Policy B. The 95% confidence interval is approximately 9.97 to 10.73 seconds, so zero improvement is not plausible. The paired estimates imply a Policy B mean near 4.11 seconds, compared with 14.46 seconds for Policy A.

Validation and Caveats

Two checks support the analysis: the fleet conservation invariant had maximum violation 0, and the depot Little's Law ratio was 0.9999, essentially 1.0. The main caveat is that this is a steady-state queueing recommendation; it does not price the station, installation constraints, staffing, or maintenance cost.